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Monsoon, 2024

B.Tech 3rd Semester Examination

Concrete Technology

Course Code: BCE23304T

Full Marks – 50

Time : 2 hours

The figure in the margin indicates full marks for the questions.

SECTION I

Answer all the questions

Marks: 10 x 1 = 10

- i. a. For quality control of Portland cement, the test essentially done is
 - i. Setting time ii. Soundness iii. Consistency iv. All of these
- b. For M10 grade of concrete, nos of bags of cement required for casting 1 cum of concrete is
 - i. 5 ii. 4 iii. 2 mix iv. 6
- c. Curing
 - i. Reduces the shrinkage of concrete ii. Preserves the properties of concrete iii. Prevents the loss of water by evaporation iv. All of the above
- d. If the standard deviation for M20 concrete is 4, then the target mean strength is
 - i. 26.6 N/mm² ii. 29 N/mm² iii. 20 N/mm² iv. None of the above
- e. For M20 grade of 1 cum concrete mix the quantity of cement required by volume is
 - i. 0.27 cum ii. 0.72 cum iii. 0.45 cum iv. 0.55 cum
- f. Calculate standard consistency (%) when 100 ml of water is added in 1000 g of cement.
 - i. 10 ii. 1000 iii. 100 iv. 1
- g. What is the final setting time?
 - i) the time at which cement paste loses its plasticity
 - ii) The time to reach that stage at which cement paste loses its elasticity
 - iii) The time at which cement paste gains its plasticity
 - iv) The time when cement paste becomes hardened
- h. If the strength gained by M20 concrete within 7 days is 13.5N/mm², then the percentage of strength gained is approximately

i) 67% ii) 76% iii) 50% iv) 55%

i. If the water/cement ratio is 0.55 for 50kg mass of cement, then the water content of the mix is

a) 27.5 b) 2.75 c) 275 and d) 0.275

i) water-cement ratio ii) Aggregate-cement ratio iii) Cement content iv) Water content

j. Gypsum added in concrete acts as

i) Retarder ii) Accelerator iii) Plasticizer iv) None of the above

SECTION 2

Answer any one question from this section

2.

a) What is NDT test? Explain its importance and what are the different types of materials that can be tested using NDT test? List out two differences between destructive test and non destructive test of materials. Identify the areas in which core cutting method can be effectively applied for determining the strength of concrete.

b) Explain slump test for measuring workability of concrete with appropriate diagram. (3+4+3=10)
(10)

SECTION 3

Answer any one question from this section

3.

a) Compare in tabular form the Minimum Cement Content, Maximum water-cement ratio and Minimum grade of concrete for both PCC and RCC as per IS: 456:2000 for different environmental exposure conditions.

b) Show the procedural steps for concrete mix design as per IS code of practice. (10)
(10)

SECTION 4

Answer any one question from this section:

4.

a) Estimate the quantity of all the elements such as mass of cement, total mass of aggregate, mass of fine aggregate, mass of coarse aggregate, volume of water and volume of admixtures for 1 cum of M35 grade Concrete with fly ash.

(10)

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- b) Determine the quantity of all the ingredients required for manufacturing 3.5 cum of M25 grade of concrete. Find the quantities as per their respective standard units. (10)

SECTION 5

Answer any one question from this section

5.

- a) List out a comparative analysis between how accelerators, retarders and plasticizers used in manufacturing of concrete affects the properties of concrete. (10)
- b. Low heat cement is more suitable for construction of massive hydrological concrete structures. Defend the statement in your own words. (10)
